Pandas

Pandas is a library for Data Analysis built off of NumPy.
Fantastic documentation:

- https://pandas.pydata.org/docs/



Using Pandas

- Tools for reading and writing data between many formats.
- Intelligently grab data based on indexing,logic, subsetting, and more.
- Handle missing data.
- Adjust and restructure data.

- Series and DataFrames
- Conditional Filtering and Useful Methods
- Missing Data
- Group By Operations
- Combining DataFrames
- Text Methods and Time Methods
- Inputs and Outputs

Let’s get started!

Series

Series

- A data structure in Pandas that holds an array of information along with a named index.
- The named index differentiates this from a simple NumPy array.
- **Formal Definition:** One-dimensional ndarray with axis labels

Numeric index in Numpy array

Index	Data
0	1776
1	1867
2	1821

Pandas Series adds on a labeled index

Labeled Index	Data
USA	1776
CANADA	1867
MEXICO	1821

Data is still numerically organized

Numeric Index	Labeled Index	Data
0	USA	1776
1	CANADA	1867
2	MEXICO	1821

- Let’s see how to create a Pandas Series object.
- We’ll also see some key properties and operations.
- Later on we will learn how to combine Series with a shared index to create a tabular data structure called a DataFrame.

Let’s code !!

DataFrames

- A DataFrame is a table of columns and rows in pandas that we can easily restructure and filter.
- **Formal Definition:** A group of Pandas Series objects that *share* the same index.

Example of a Series

Index	Year
USA	1776
CANADA	1867
MEXICO	1821

Example of Series with Same Index

Index	Year
USA	1776
CANADA	1867
MEXICO	1821

Index	Pop
USA	328
CANADA	38
MEXICO	126

Index	GDP
USA	20.5
CANADA	1.7
MEXICO	1.22

Example of Series with Same Index

Index	Year
USA	1776
CANADA	1867
MEXICO	1821

Index	Pop
USA	328
CANADA	38
MEXICO	126

Index	GDP
USA	20.5
CANADA	1.7
MEXICO	1.22

Index	Year	Pop	GDP
USA	1776	328	20.5
CANADA	1867	38	1.7
MEXICO	1821	126	1.22

DataFrame is the main Pandas object we will work with and it is **extremely** useful!